

Submission PHYS1-MID-SUB06

Question PHYS1-MID-Q01

notation changes halfway:

1. $v \approx v_{\text{initial}} + at \approx 4 + 3(5) \approx 19 \text{ m/s}$
2. $\Delta x \approx v_{\text{initial}}t + \frac{1}{2}at^2 \approx 4(5) + \frac{1}{2}(3)(5)^2 \approx 57.5 \text{ m}$

marks, subscripts, and units are mixed up

result written as: $v=19 \text{ m/s}$; displacement=57.5 m (please interpret)

Question PHYS1-MID-Q02

notation changes halfway:

1. $N \approx mg \approx 58.8 \text{ N}$; friction $\approx \mu N \approx 11.8 \text{ N}$
2. $\Sigma F \approx 17 - 11.8 \approx 5.2 \text{ N}$
3. $a \approx \Sigma F/m \approx 0.87 \text{ m/s}^2$

marks, subscripts, and units are mixed up

result written as: $a=0.87 \text{ m/s}^2$; friction opposes motion (please interpret)

Question PHYS1-MID-Q03

notation changes halfway:

1. $mgh \approx \frac{1}{2}mv^2 \Rightarrow v \approx \sqrt{2gh} \approx \sqrt{2 \cdot 9.8 \cdot 8}$
2. $v \approx 12.52 \text{ m/s}$; mass cancels

marks, subscripts, and units are mixed up

result written as: $v=12.52 \text{ m/s}$ (please interpret)

Question PHYS1-MID-Q04

notation changes halfway:

1. $p_{\text{total, before}} \approx p_{\text{total, after}}$
2. It applies when external impulse on the system is negligible.

marks, subscripts, and units are mixed up

result written as: total momentum before equals after when external impulse is negligible (please interpret)

Question PHYS1-MID-Q05

notation changes halfway:

1. Impulse is area under $F(t)$: $J \approx F\Delta t \approx 6(0.5) \approx 3 \text{ N}\cdot\text{s}$.
2. $\Delta p \approx J$, so momentum changes by $3 \text{ kg}\cdot\text{m/s}$ in the force direction.

marks, subscripts, and units are mixed up

result written as: impulse=3 N·s; momentum changes by $3 \text{ kg}\cdot\text{m/s}$ (please interpret)

Question PHYS1-MID-Q06

notation changes halfway:

1. $v \approx v_{\text{initial}} + at \approx 7 + 3(2) \approx 13 \text{ m/s}$
2. $\Delta x \approx v_{\text{initial}}t + \frac{1}{2}at^2 \approx 7(2) + \frac{1}{2}(3)(2)^2 \approx 20 \text{ m}$

marks, subscripts, and units are mixed up

result written as: $v=13 \text{ m/s}$; displacement=20 m (please interpret)

Question PHYS1-MID-Q07

notation changes halfway:

1. $N \approx mg \approx 19.6 \text{ N}$; friction $\approx \mu N \approx 3.9 \text{ N}$

2. $\Sigma F \approx 16 - 3.9 \approx 12.1 \text{ N}$

3. $a \approx \Sigma F/m \approx 6.04 \text{ m/s}^2$

marks, subscripts, and units are mixed up

result written as: $a=6.04 \text{ m/s}^2$; friction opposes motion (please interpret)

Question PHYS1-MID-Q08

notation changes halfway:

1. $mgh \approx \frac{1}{2}mv^2 \Rightarrow v \approx \sqrt{2gh} \approx \sqrt{2 \cdot 9.8 \cdot 9}$

2. $v \approx 13.28 \text{ m/s}$; mass cancels

marks, subscripts, and units are mixed up

result written as: $v=13.28 \text{ m/s}$ (please interpret)